

46-921 Supplement: Proofs

These proofs will not be covered on the homeworks or the exam, but it is important to know the claims.

Throughout assume that $X_1, X_2, \dots, X_n$ are i.i.d. from the normal distribution with mean μ and variance σ^2 .

Claim 1: $\bar{X}$ and S^2 are independent.

Proof: First, note that since $X_1, X_2, \dots, X_n$ are independent and normal, then the random vector $\mathbf{X}$ consisting of $X_1, X_2, \dots, X_n$ is multivariate normal with covariance matrix $\Sigma = \sigma^2 \mathbf{I}$. (To see this, derive the joint pdf of $\mathbf{X}$ by taking the product of the pdfs of $X_1, X_2, \dots, X_n$ and note that its form matches the multivariate normal with covariance matrix Σ .)

Fix any $i \in \{1, 2, \dots, n\}$. We see that we can write

$$\begin{bmatrix} \bar{X} \\ (X_i - \bar{X}) \end{bmatrix} = \mathbf{A}_i \mathbf{X}$$

with

$$\mathbf{A}_i = \begin{bmatrix} \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \dots & \frac{1}{n} \\ -\frac{1}{n} & \dots & -\frac{1}{n} & 1 - \frac{1}{n} & -\frac{1}{n} & \dots & -\frac{1}{n} \end{bmatrix}$$

where the $1 - 1/n$ entry in the second row of $\mathbf{A}_i$ is in the i^{th} column. This implies that

$$\begin{bmatrix} \bar{X} \\ (X_i - \bar{X}) \end{bmatrix}$$

is bivariate normal with covariance matrix $\mathbf{A}_i \Sigma \mathbf{A}_i^T = \sigma^2 \mathbf{A}_i \mathbf{A}_i^T$. Finally, the off-diagonal entry of $\sigma^2 \mathbf{A}_i \mathbf{A}_i^T$ is

$$\sigma^2 \left[\left(\frac{1}{n} \right) \left(1 - \frac{1}{n} \right) + \sum_{j \neq i} \left(\frac{1}{n} \right) \left(-\frac{1}{n} \right) \right] = 0.$$

Of course, this off-diagonal element is the covariance between $\bar{X}$ and $(X_i - \bar{X})$. The conclusion is that $\bar{X}$ and $(X_i - \bar{X})$ are uncorrelated and hence independent.

Next, note that the sample variance S^2 can be written as a function of $(X_i - \bar{X})$ for $i = 1, 2, \dots, n$:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Hence, S^2 is a function of random variables, each of which is independent of $\bar{X}$. This implies that $\bar{X}$ is independent of S^2 . $\square$

Claim 2: The quantity

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has the standard normal distribution.

Proof: We showed this in Probability. $\square$

Claim 3: The quantity

$$\frac{(n-1)S^2}{\sigma^2}$$

has the chi-squared distribution with $n-1$ degrees of freedom.

Proof: To start, note that

$$\begin{aligned} \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 &= \sum_{i=1}^n \left(\frac{X_i - \bar{X} + \bar{X} - \mu}{\sigma} \right)^2 \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n \left[(X_i - \bar{X})^2 + 2(X_i - \bar{X})(\bar{X} - \mu) + (\bar{X} - \mu)^2 \right] \\ &= \frac{1}{\sigma^2} \left[\sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2 \right] \quad \text{since } \sum_i (X_i - \bar{X}) = 0 \\ &= \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2 \\ &= W_2 + W_3 \end{aligned}$$

where

$$W_2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \quad \text{and} \quad W_3 = \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2.$$

Also define

$$W_1 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2$$

so that we have shown $W_1 = W_2 + W_3$.

We know that W_1 has the chi-squared distribution with n degrees of freedom and that W_3 has the chi-squared distribution with a single degree of freedom. Note that the only random variable W_2 depends on is S^2 and the only random variable W_3 depends on is $\bar{X}$. Since S^2 and $\bar{X}$ are independent, it follows that W_2 and W_3 are independent. Hence,

$$M_{W_1}(t) = M_{W_2}(t)M_{W_3}(t)$$

and

$$\begin{aligned} M_{W_2}(t) &= M_{W_1}(t) / M_{W_3}(t) \\ &= \left(\frac{1}{1-2t} \right)^{n/2} / \left(\frac{1}{1-2t} \right)^{1/2} \\ &= \left(\frac{1}{1-2t} \right)^{\left(\frac{n-1}{2}\right)}. \end{aligned}$$

So, by the uniqueness of the MGF, W_2 has the chi-squared distribution with $n - 1$ degrees of freedom. Finally,

$$W_2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 = \frac{(n-1) S^2}{\sigma^2}. \quad \square$$

Claim 4: The quantity

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}$$

has the t -distribution with $n - 1$ degrees of freedom.

Proof: Start by defining random variables

$$Z = \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right) \quad \text{and} \quad V = \frac{(n-1) S^2}{\sigma^2}.$$

Based on the above results, we know that Z has the standard normal distribution and V has the chi-squared distribution with $n - 1$ degrees of freedom. We also know that Z and V are independent.

It is not difficult to see that

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{Z}{\sqrt{V/(n-1)}}$$

and thus by the definition of the t -distribution, the claim follows. $\square$